

This Month's DVD Contents

This month's DVD contains some interesting software starting from LabVIEW and Ngspice in the electronics design section to a circuit design tool for printed circuit boards (PCBs). It also contains utility tools supporting electronics design and simulation processes

SNEHA AMBASTHA

LabVIEW

LabVIEW is a system design software that provides comprehensive tools to build any control or measurement application in minimum time. It helps in accelerating productivity, solving problems and continuing innovations with accelerated results. It also allows the user to save, present and share the results of measurements for future use. This student version of the software has limited features; however, other full versions are also available from National Instruments (NI).

LTSpice IV

This free Spice simulator with high-performance simulation technology from Linear Technologies comes with a schematic capture, waveform viewer with models and enhancements to make the simulation procedure of switching regulators easy and extremely fast.

Ngspice

Ngspice is an open source Spice circuit simulator with its code based on three open source software packages, namely, Ciderb1, Xspice and Spice3f5. This general-purpose mixed-signal or mixed-level simulation tool works for both linear and non-linear circuit analysis, that is, it can work with circuits containing both active and passive components.

VeroDes

VeroDes is a circuit design software for Veroboard or Stripboard. Converting schematics with copper tracks on Veroboard is quite difficult because these are fixed. VeroDes makes it easy to do so. It works on Linux platform and allows the user to print lay-

outs so designed on both sides of the board. It helps cut out tracks and place components at the required places, eliminating the effort required for scribbling diagrams on graph paper.

SAEAUT SCADA

SAEAUT SCADA version 5.03.0.3 is a supervisory control and data acquisition (SCADA) software that allows the user to monitor, control and store data, and process the alarm. It makes communication with various devices easy and enables database communication through dynamic data exchange (DDE) servers and open platform communications (OPC).

Advanced Trigonometry Calculator

Advanced Trigonometry Calculator version 1.8.0 is a software solution that provides a reliable means for performing mathematical calculations with less effort and with a high level of precision. It is freely available and can be used on Windows operating system (OS). It only works with CMD interface and thus requires basic knowledge of the CMD interface.

ZCad

ZCad version 0.9.8 helps create complex computer aided design (CAD) projects. Its console already contains some examples and input commands for assistance. It is a free program that runs on Windows OS.

Capture Argo

Capture Argo version 21.0.12 is a lighting application for professionals. It helps create complex lighting systems for theatres and concerts. It creates a user-friendly working

Popular resources

- 7Zip. Version v.9.38 (file archiver)
- Java Development Kit (JDK). Version 8.0.450.15 (development kit to create Java runtime environment for various applications)
- VLC. Version 2.2.1 (media player)
- Opera. Version 29.0.1795.60 (Web browser)
- Free Download Manager. Version 3.9.5 (file download manager)
- Thunderbird. Version 31.1.0 (email application)
- Mozilla Firefox. Version 38.0.1 (Web browser)
- Apache Open Office. Version 4.1.1 (open office)
- Ubuntu. Version 15.04 (operating system)

environment and allows technicians to use their knowledge and skill to create a lighting system for stage shows. It requires the knowledge of light physics and human perception for creating modern lighting systems.

Cimg

Cimg is an open source library and a modern C++ toolkit for image processing. It allows the user to manage images using C++ codes. It can not only load or save image file formats but also access pixel values and draw primitive images. It is a highly portable and lightweight software and is thread-safe. It is compatible with compilers like Visual C++, g++ and clang++.

TarsosDSP

Being a Java library for audio processing, TarsosDSP aims to provide an interface for the implementation of music algorithms in pure Java without external dependencies. The Java library demonstrates the working of the digital-signal processing (DSP) algorithm. It has a number of pitch-detection algorithms and a percussion-onset detector. ●

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